

Taylor and Maclaurin Series



The Taylor series formula

- 1 State the general Taylor series formula for $f(x)$ centered at $x = a$.
 - 2 Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x - 1)^3$ term.
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Common Maclaurin series

- 3 State the Maclaurin series for e^x , $\sin x$, and $\cos x$.
 - 4 Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2} .
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Using known series to build new series

- 5 Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x \sin x$.
 - 6 Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.
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The Lagrange error bound

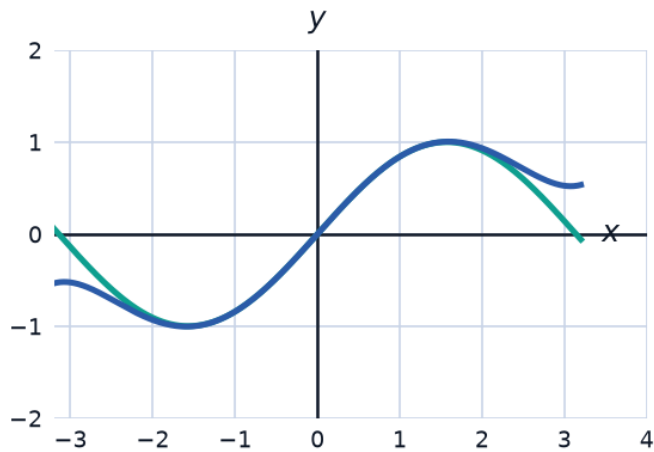
- 7 State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n .
 - 8 Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \leq 1$ for all z .
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Radius of convergence of Taylor series

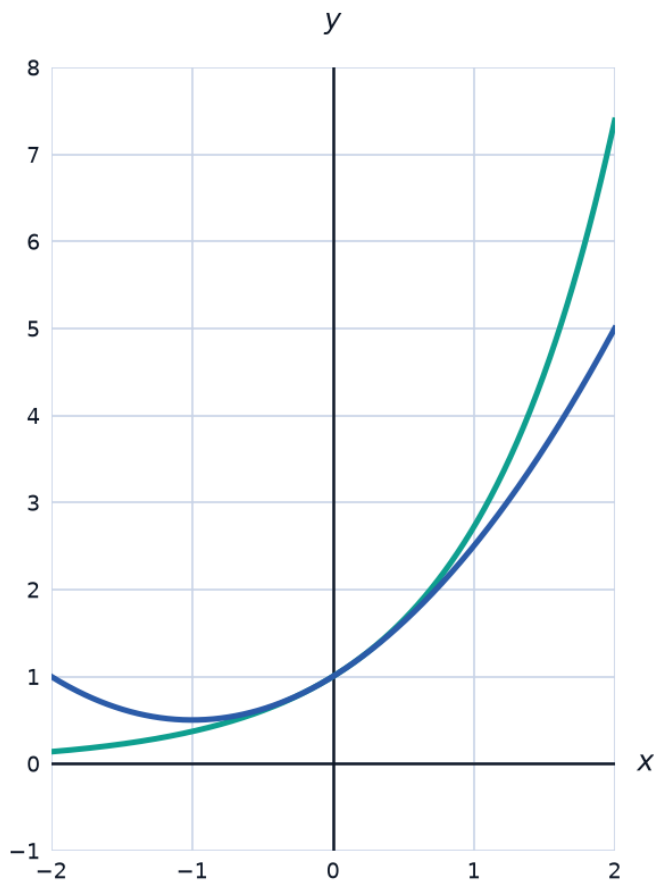
- 9 State the radius of convergence of the Maclaurin series for e^x , $\sin x$, and $\cos x$.
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Visualizing Taylor polynomial approximations

- 10 Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.



- 11 Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x .



Building Taylor series at a nonzero center

- 12 Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.
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Applying the Lagrange Error Bound to bound accuracy requirements

- 13 Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \leq 1$, find the maximum error in approximating $\cos(0.3)$.
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The binomial series

- 14 State the binomial series for $(1+x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1+x}$ (i.e. $k = \frac{1}{2}$) up through the quadratic term.
- 15 Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.
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Deriving new Maclaurin series from known ones

- 16 Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.
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Multiple choice — easy 15

- 17 Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^5 .

- a. $\frac{121}{40}$
- b. $\frac{81}{8}$
- c. $\frac{1}{40}$
- d. $\frac{81}{40}$

- 18 Using $e^{7x} = \sum_{n=0}^{\infty} \frac{(7x)^n}{n!}$, find the coefficient of x^4 .

- a. $\frac{7}{24}$
- b. $\frac{2425}{24}$
- c. $\frac{2401}{6}$
- d. $\frac{2401}{24}$

19 Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^2 .

a. 18

b. 19

c. 3

d. 36

20 Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{11}{2}$

b. $\frac{27}{2}$

c. $\frac{1}{2}$

d. $\frac{9}{2}$

21 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^4 .

a. $\frac{512}{3}$

b. $\frac{2048}{3}$

c. $\frac{515}{3}$

d. $\frac{1}{3}$

22 Using $e^{4x} = \sum_{n=0}^{\infty} \frac{(4x)^n}{n!}$, find the coefficient of x^5 .

a. $\frac{143}{15}$

b. $\frac{128}{15}$

c. $\frac{1}{30}$

d. $\frac{128}{3}$

23 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{4}{3}$

b. 256

c. $\frac{259}{3}$

d. $\frac{256}{3}$

24 Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^2 .

a. $\frac{11}{2}$

b. $\frac{9}{2}$

c. 9

d. $\frac{3}{2}$

25 Using $e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$, find the coefficient of x^2 .

a. 4

b. 3

c. 2

d. 1

26 Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^3 .

a. 36

b. 108

c. 1

d. 37

27 Using $e^{4x} = \sum_{n=0}^{\infty} \frac{(4x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{2}{3}$

b. 32

c. $\frac{35}{3}$

d. $\frac{32}{3}$

28 Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^5 .

a. 324

b. $\frac{324}{5}$

c. $\frac{329}{5}$

d. $\frac{1}{20}$

29 Using $e^{9x} = \sum_{n=0}^{\infty} \frac{(9x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{729}{2}$

b. $\frac{243}{2}$

c. $\frac{245}{2}$

d. $\frac{3}{2}$

30 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^2 .

a. 32

b. 64

c. 33

d. 4

31 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^5 .

a. $\frac{1}{15}$

b. $\frac{4096}{3}$

c. $\frac{4111}{15}$

d. $\frac{4096}{15}$

Multiple choice — medium 15

32 Using $\sin(6x) = \sum_{n=0}^{\infty} \frac{(-1)^n (6x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. -35

b. 36

c. -36

d. 108

33 Using $\sin(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $\frac{19}{15}$

b. $\frac{4}{3}$

c. $-\frac{4}{15}$

d. $\frac{4}{15}$

34 Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $\frac{125}{2}$

b. $-\frac{125}{6}$

c. $\frac{125}{6}$

d. $-\frac{119}{6}$

35 Using $\sin(9x) = \sum_{n=0}^{\infty} \frac{(-1)^n (9x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $\frac{19683}{8}$

b. $-\frac{19683}{40}$

c. $\frac{19723}{40}$

d. $\frac{19683}{40}$

36 Using $\sin(4x) = \sum_{n=0}^{\infty} \frac{(-1)^n (4x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $-\frac{29}{3}$

b. $\frac{32}{3}$

c. 32

d. $-\frac{32}{3}$

37 Using $\sin(7x) = \sum_{n=0}^{\infty} \frac{(-1)^n (7x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $-\frac{337}{6}$

b. $-\frac{343}{6}$

c. $\frac{343}{6}$

d. $\frac{343}{2}$

38 Using $\sin(4x) = \sum_{n=0}^{\infty} \frac{(-1)^n (4x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $\frac{128}{15}$

b. $\frac{143}{15}$

c. $-\frac{128}{15}$

d. $\frac{128}{3}$

39 Using $\sin(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. 4

b. $-\frac{1}{3}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

40 Using $\sin(6x) = \sum_{n=0}^{\infty} \frac{(-1)^n (6x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $\frac{324}{5}$

b. $-\frac{324}{5}$

c. $\frac{329}{5}$

d. 324

41 Using $\sin(9x) = \sum_{n=0}^{\infty} \frac{(-1)^n (9x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $-\frac{241}{2}$

b. $-\frac{243}{2}$

c. $\frac{729}{2}$

d. $\frac{243}{2}$

42 Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $-\frac{81}{40}$

b. $\frac{81}{8}$

c. $\frac{121}{40}$

d. $\frac{81}{40}$

43 Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $\frac{9}{2}$

b. $\frac{27}{2}$

c. $-\frac{7}{2}$

d. $-\frac{9}{2}$

44 Using $\sin(8x) = \sum_{n=0}^{\infty} \frac{(-1)^n (8x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $\frac{256}{3}$

b. $-\frac{253}{3}$

c. $-\frac{256}{3}$

d. 256

45 Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $-\frac{625}{24}$

b. $\frac{649}{24}$

c. $\frac{625}{24}$

d. $\frac{3125}{24}$

46 Using $\sin(8x) = \sum_{n=0}^{\infty} \frac{(-1)^n (8x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $-\frac{4096}{15}$

b. $\frac{4096}{3}$

c. $\frac{4096}{15}$

d. $\frac{4111}{15}$

Multiple choice — hard 15

- 47 Use the Maclaurin series for e^u (with $u = 3x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{3x^2} .
- a. $\frac{2187}{8}$
 - b. $\frac{9}{4480}$
 - c. $\frac{35}{8}$
 - d. $\frac{27}{8}$
- 48 Use the Maclaurin series for e^u (with $u = 4x^2$) to find the coefficient of x^6 in the Maclaurin series for e^{4x^2} .
- a. $\frac{32}{3}$
 - b. $\frac{2048}{3}$
 - c. $\frac{4}{45}$
 - d. $\frac{35}{3}$
- 49 Use the Maclaurin series for e^u (with $u = 2x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{2x^2} .
- a. $\frac{2}{3}$
 - b. $\frac{1}{2520}$
 - c. $\frac{5}{3}$
 - d. $\frac{32}{3}$
- 50 Use the Maclaurin series for e^u (with $u = 9x^2$) to find the coefficient of x^6 in the Maclaurin series for e^{9x^2} .
- a. $\frac{245}{2}$
 - b. $\frac{81}{80}$
 - c. $\frac{243}{2}$
 - d. $\frac{177147}{2}$

- 51 Use the Maclaurin series for e^u (with $u = 7x^2$) to find the coefficient of x^4 in the Maclaurin series for e^{7x^2} .
- a. $\frac{49}{24}$
 - b. $\frac{2401}{2}$
 - c. $\frac{49}{2}$
 - d. $\frac{51}{2}$
- 52 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 6$, $n = 2$, and $|x-a| = 1$, find the error bound.
- a. 2
 - b. $\frac{7}{6}$
 - c. 3
 - d. 1
- 53 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 9$, $n = 2$, and $|x-a| = 1$, find the error bound.
- a. $\frac{5}{3}$
 - b. $\frac{3}{2}$
 - c. $\frac{5}{2}$
 - d. $\frac{9}{2}$
- 54 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 5$, $n = 3$, and $|x-a| = 1$, find the error bound.
- a. $\frac{5}{6}$
 - b. $\frac{5}{24}$
 - c. $\frac{29}{24}$
 - d. $\frac{1}{4}$

- 55 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$ with $M = 5$, $n = 4$, and $|x - a| = 1$, find the error bound.
- $\frac{1}{24}$
 - $\frac{5}{24}$
 - $\frac{1}{20}$
 - $\frac{25}{24}$
- 56 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$ with $M = 4$, $n = 2$, and $|x - a| = 1$, find the error bound.
- $\frac{5}{3}$
 - 2
 - $\frac{5}{6}$
 - $\frac{2}{3}$
- 57 Using $\cosh(2x) = \frac{e^{2x} + e^{-2x}}{2} = \sum_{n=0}^{\infty} \frac{(2x)^{2n}}{(2n)!}$, find the coefficient of x^2 .
- 4
 - $\frac{2}{3}$
 - 2
 - 3
- 58 Using $\cosh(8x) = \frac{e^{8x} + e^{-8x}}{2} = \sum_{n=0}^{\infty} \frac{(8x)^{2n}}{(2n)!}$, find the coefficient of x^6 .
- $\frac{32768}{15}$
 - $\frac{16384}{315}$
 - $\frac{16384}{45}$
 - $\frac{16429}{45}$

59 Using $\cosh(6x) = \frac{e^{6x} + e^{-6x}}{2} = \sum_{n=0}^{\infty} \frac{(6x)^{2n}}{(2n)!}$, find the coefficient of x^2 .

a. 18

b. 36

c. 6

d. 19

60 Using $\cosh(4x) = \frac{e^{4x} + e^{-4x}}{2} = \sum_{n=0}^{\infty} \frac{(4x)^{2n}}{(2n)!}$, find the coefficient of x^4 .

a. $\frac{35}{3}$

b. $\frac{128}{3}$

c. $\frac{32}{3}$

d. $\frac{32}{15}$

61 Using $\cosh(7x) = \frac{e^{7x} + e^{-7x}}{2} = \sum_{n=0}^{\infty} \frac{(7x)^{2n}}{(2n)!}$, find the coefficient of x^6 .

a. $\frac{118369}{720}$

b. $\frac{117649}{120}$

c. $\frac{16807}{720}$

d. $\frac{117649}{720}$